Simulation-based verification of bfpwr

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Package version 0.3
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This report summarizes the simulation-based verification checks for **bfpwr**. The report focuses on the verification settings and results: the simulated designs, analysis priors, evidence thresholds, sampling schedules, and the size of any package calculation differences relative to the simulation targets.

1 z-Test

Simulated Design Priors

Table 1: Simulated design priors for the z-Test verification run. The Design column gives short IDs used by the analysis-prior table.

Design	Design Prior	Sequence	Repetitions	USD
Z1	Point(0)	10 to 500 by 10	10000	1
Z2	Point(0.2)	10 to 500 by 10	10000	1
Z3	Point(0.5)	10 to 500 by 10	10000	1
Z4	Normal(0.5, 0.1)	10 to 500 by 10	10000	1
Z5	Normal(0.2, 0.3)	10 to 500 by 10	10000	1
Z6	Normal(0.2, 0.5)	10 to 500 by 10	10000	1
Z7	Normal(0, 0.5)	10 to 500 by 10	10000	1
Z8	Point(-0.3)	10 to 500 by 10	10000	1
Z9	Point(0)	10 to 500 by 10	10000	50
Z10	Point(10)	10 to 500 by 10	10000	50
Z11	Normal(10, 20)	10 to 500 by 10	10000	50
Z12	Point(0)	10 to 10000 by 10	10000	1
Z13	Point(0.2)	10 to 10000 by 10	10000	1
Z14	Normal(0.2, 0.1)	10 to 10000 by 10	10000	1
Z15	Normal(0.2, 0.5)	10 to 10000 by 10	10000	1

Analysis Priors

Table 2: Analysis priors used for the z-Test verification run. Designs lists the matching short IDs from the design-prior table.

Function	H1	H0	Designs
bf01	Normal(0, 0.707)	Point(0)	Z1-Z8, Z12-Z15
bf01	Normal(0, 20)	Point(0)	Z9-Z11
bf01	Normal(0.2, 0.3)	Point(0)	Z1-Z8, Z12-Z15

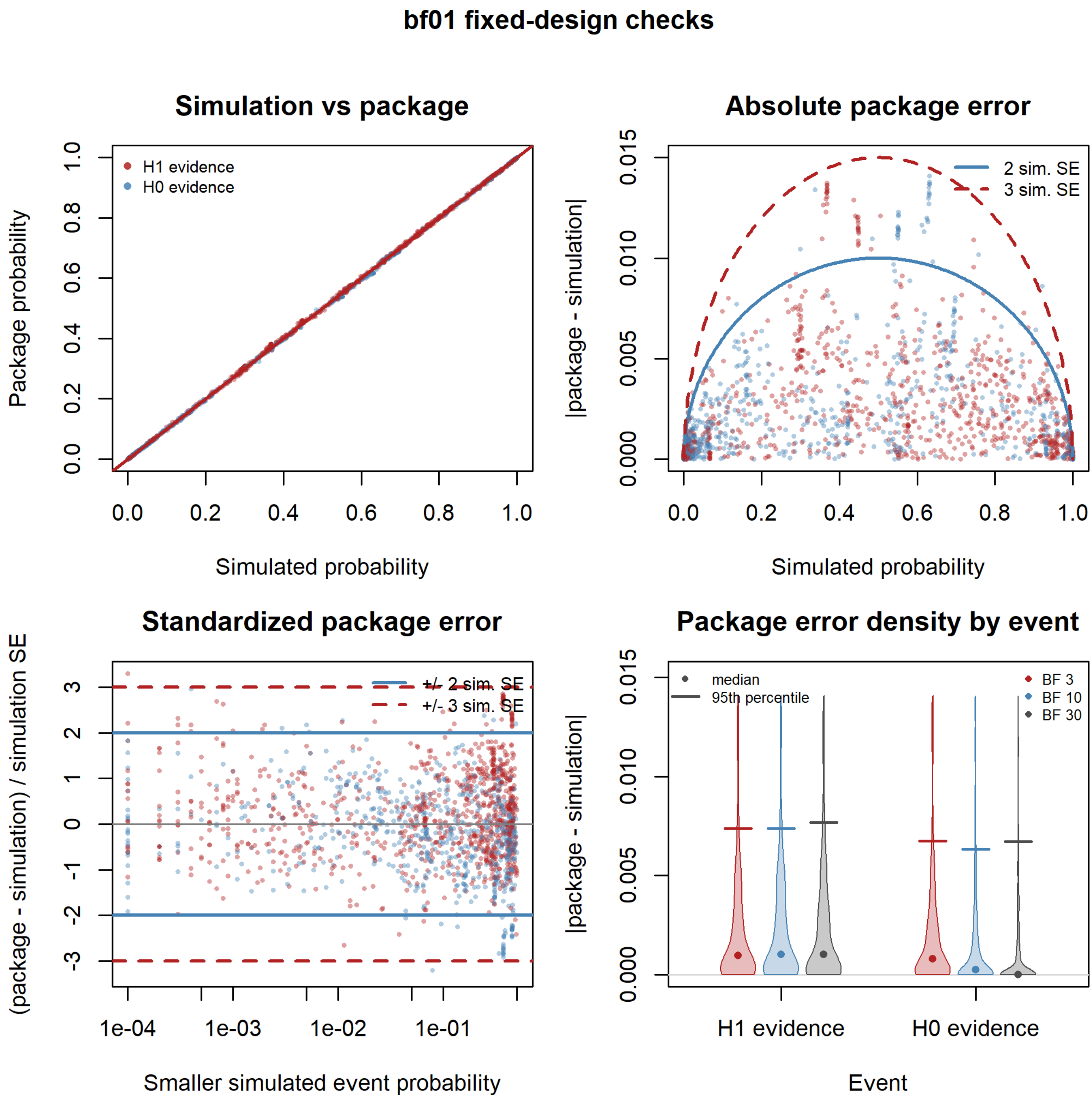
bf01	Normal(0.2, 0.707)	Point(0.2)	Z2-Z6, Z13-Z15
bf01	Normal(10, 20)	Point(0)	Z9-Z11
bf01	Point(0.2)	Point(0)	Z1-Z8, Z12-Z15
bf01	Point(0.5)	Point(0)	Z1-Z8, Z12-Z15
bf01	Point(0.5)	Point(0.2)	Z2-Z6, Z13-Z15
bf01	Point(10)	Point(0)	Z9-Z11
dirbf01	Normal$_+$(0, 0.707; > 0)	Normal$_-$(0, 0.707; ≤ 0)	Z1-Z8, Z12-Z15
dirbf01	Normal$_+$(0, 20; > 0)	Normal$_-$(0, 20; ≤ 0)	Z9-Z11
dirbf01	Normal$_+$(0.2, 0.707; > 0.2)	Normal$_-$(0.2, 0.707; ≤ 0.2)	Z2-Z6, Z13-Z15
nmbf01	Moment(0.354)	Point(0)	Z1-Z8, Z12-Z15
nmbf01	Moment(0.354)	Point(0.2)	Z2-Z6, Z13-Z15
nmbf01	Moment(0.707)	Point(0)	Z1-Z8, Z12-Z15
nmbf01	Moment(7.071)	Point(0)	Z9-Z11

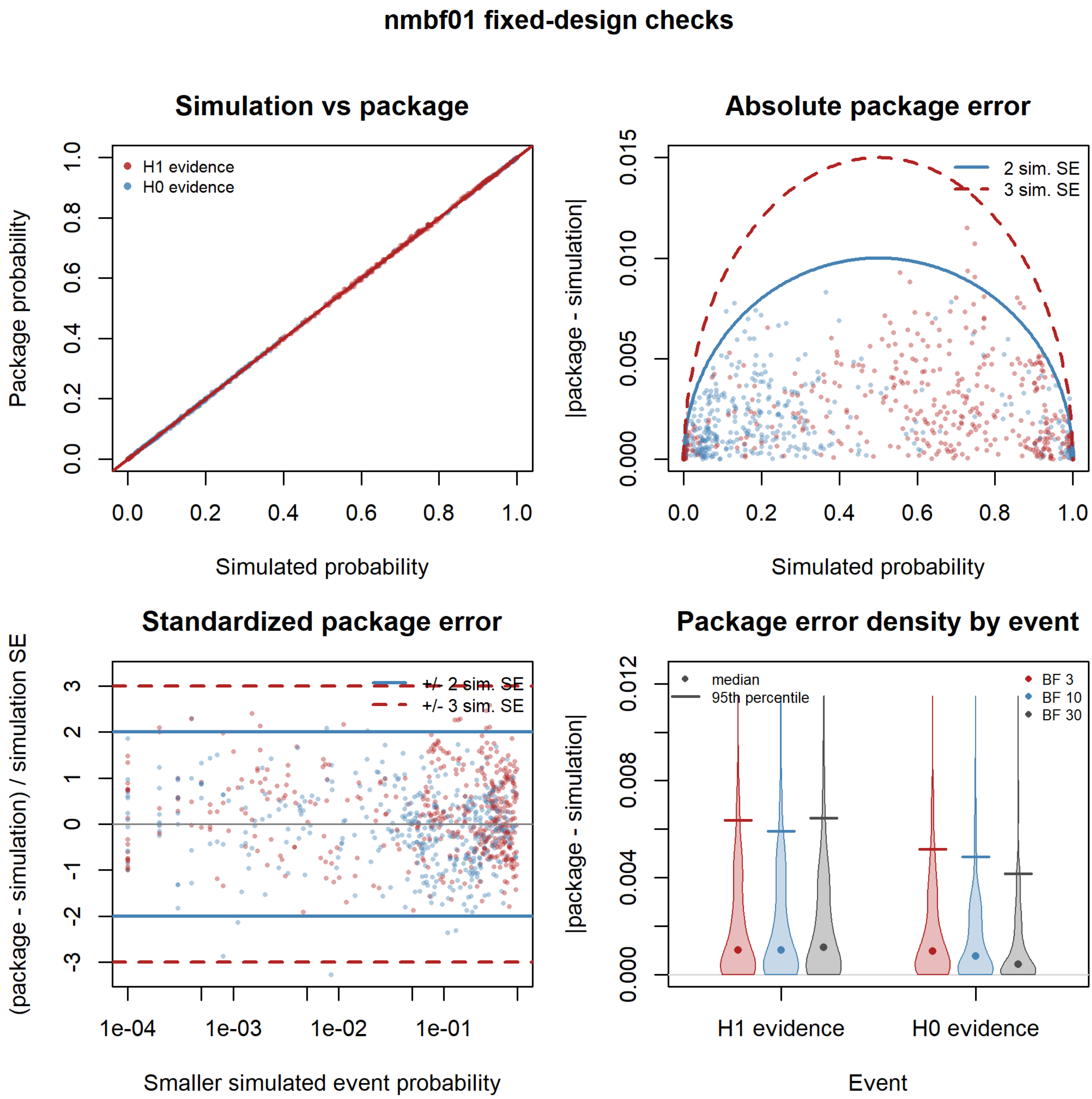
1.1 Fixed-Design

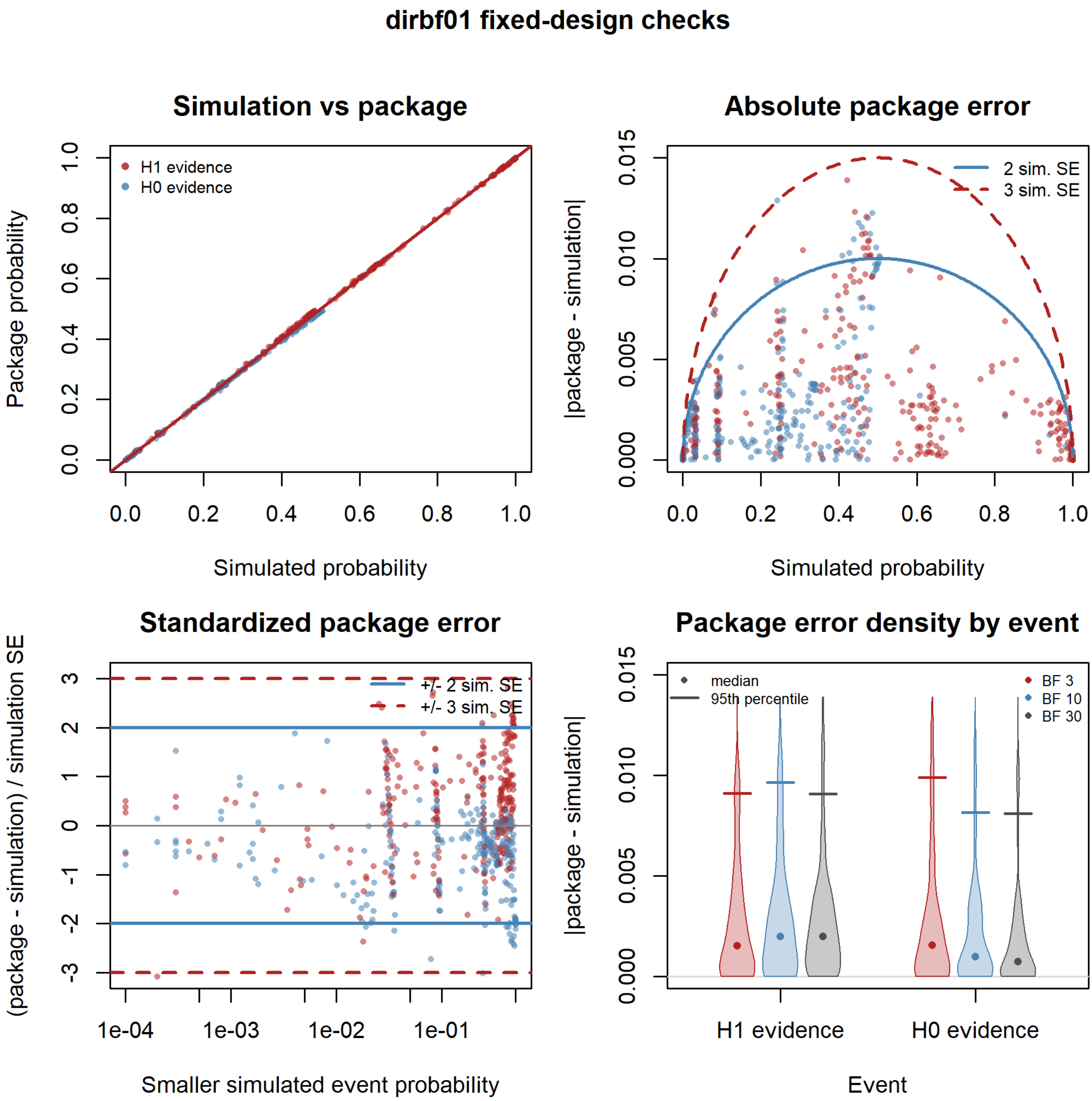
Table 3: Fixed-design verification settings for z-Test analyses.

Function	Analysis priors	Evidence thresholds
bf01	73	BF 3, 10, 30
nmbf01	35	BF 3, 10, 30
dirbf01	23	BF 3, 10, 30

The fixed-design plots compare package probability calculations with simulated event frequencies. Some expensive checks are plotted from a precomputed subset; the settings table above describes the full verification design. The panels compare simulated and package probabilities, absolute error, standardized error, and error distributions by event.


bf01 fixed-design checks
Simulation vs package
H1 evidence
H0 evidence
Package probability
Simulated probability
Absolute package error
2 sim. SE
3 sim. SE
|package - simulation|
Simulated probability
Standardized package error
+/- 2 sim. SE
+/- 3 sim. SE
(package - simulation) / simulation SE
Smaller simulated event probability
Package error density by event
median
95th percentile
BF 3
BF 10
BF 30
|package - simulation|
H1 evidence
H0 evidence
Event


nmbf01 fixed-design checks
Simulation vs package
H1 evidence
H0 evidence
Package probability
Simulated probability
Absolute package error
2 sim. SE
3 sim. SE
|package - simulation|
Simulated probability
Standardized package error
+/- 2 sim. SE
+/- 3 sim. SE
(package - simulation) / simulation SE
Smaller simulated event probability
Package error density by event
median
95th percentile
BF 3
BF 10
BF 30
|package - simulation|
H1 evidence
H0 evidence
Event



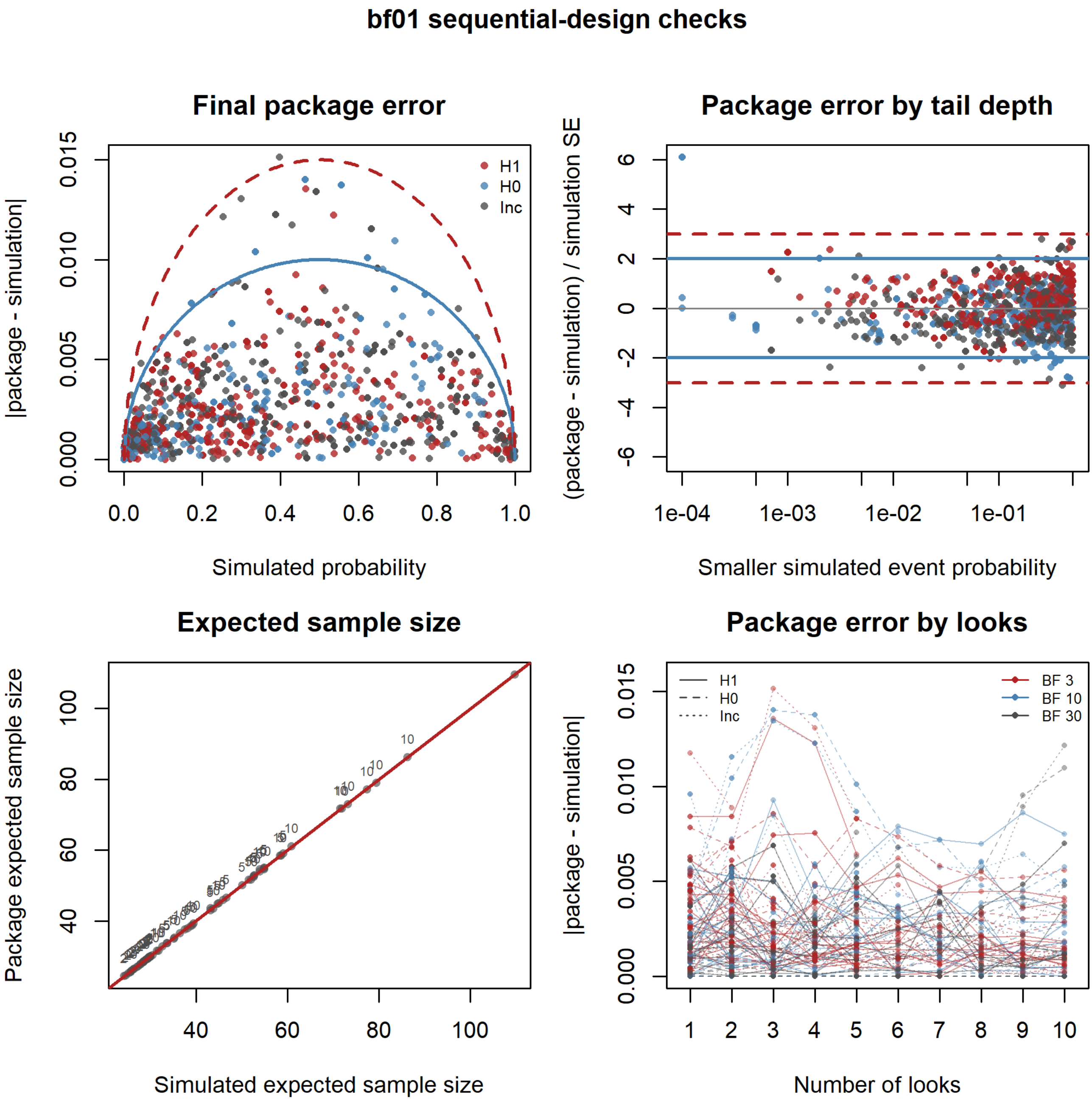
1.2 Sequential-Design

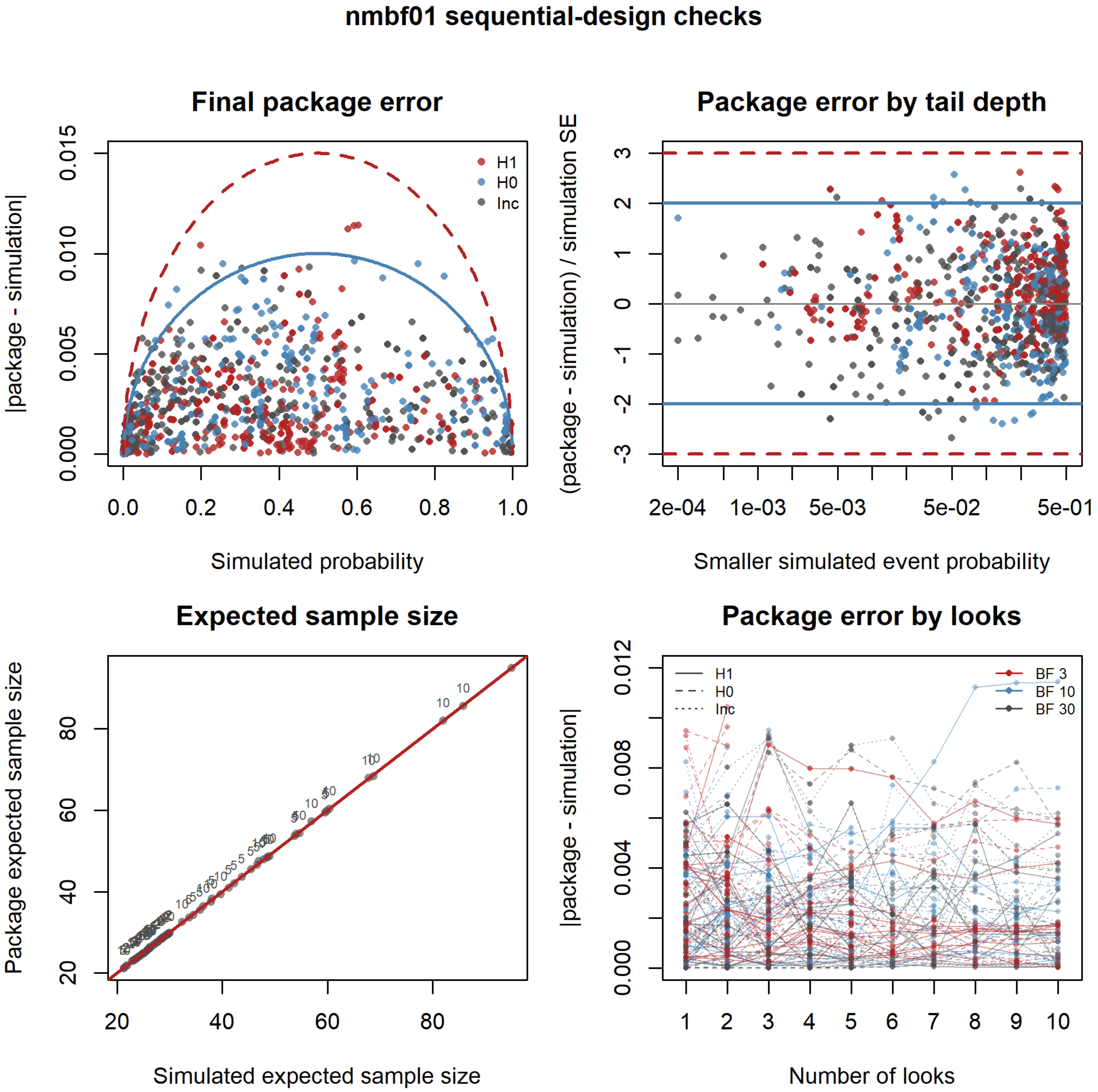
Sequential schedules use looks every 10 observations: one family starts at $n = 20$ and includes 2–20 looks; the stress schedules start at $n = 10$ and include 50, 100, or 200 looks. The 100- and 200-look schedules are compatible with the longer design sequence only.

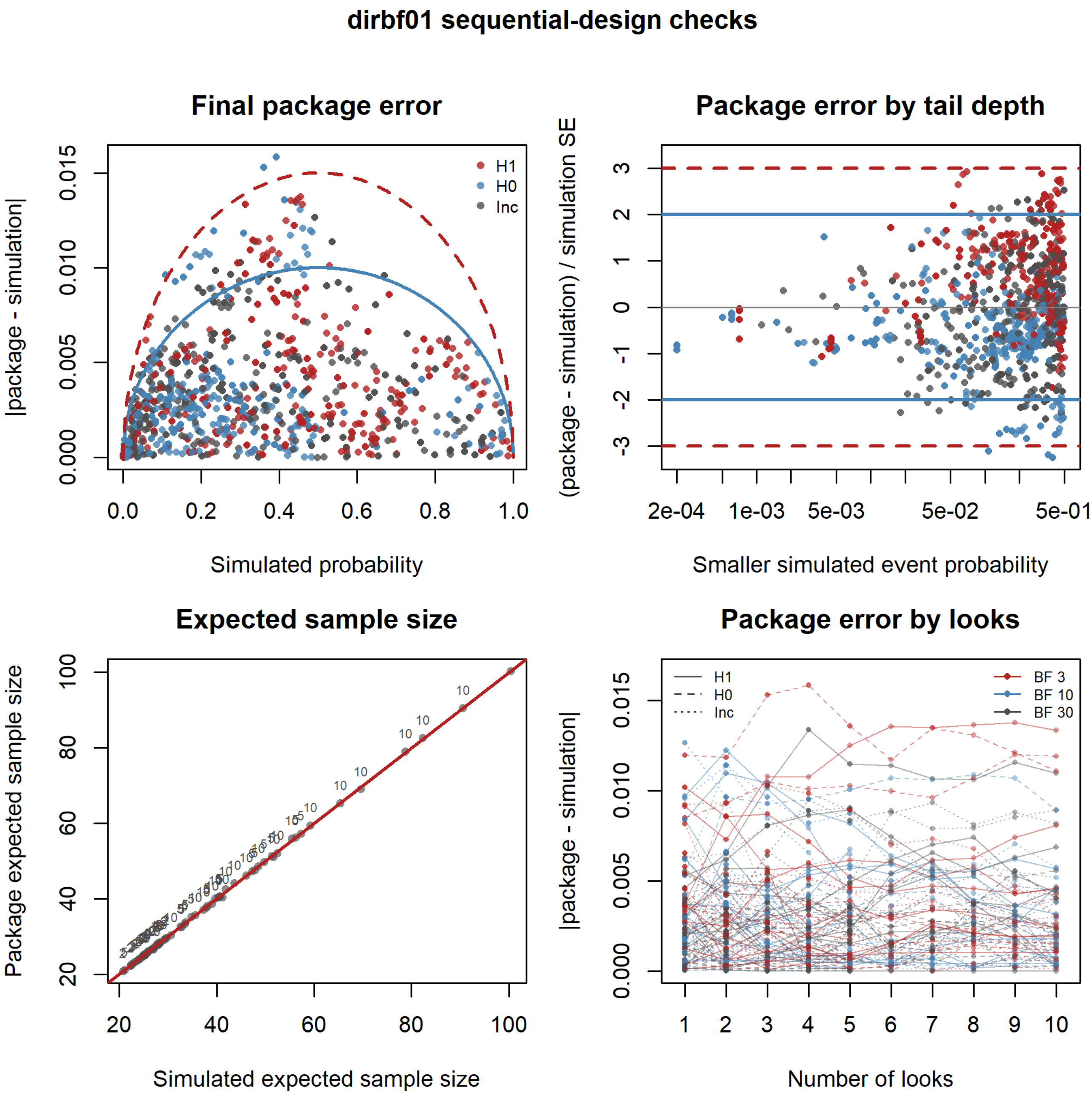
Table 4: Sequential-design verification settings for z-Test analyses.

Function	Analysis priors	Evidence thresholds	Look sequence
bf01	73	BF 3, 10, 30	start 20, by 10 (2-20 looks); start 10, by 10 (50, 100, 200 looks)
nmbf01	35	BF 3, 10, 30	start 20, by 10 (2-20 looks); start 10, by 10 (50, 100, 200 looks)
dirbf01	23	BF 3, 10, 30	start 20, by 10 (2-20 looks); start 10, by 10 (50, 100, 200 looks)

Sequential-design plots compare package probability and expected-sample-size calculations with the selected z-test simulation settings. The panels show final package error, package error by tail depth, expected sample size agreement, and design-level package-error traces by number of looks.




nmbf01 sequential-design checks
Final package error
H1
H0
Inc
|package - simulation|
Simulated probability
Package error by tail depth
(package - simulation) / simulation SE
Smaller simulated event probability
Expected sample size
Package expected sample size
Simulated expected sample size
Package error by looks
H1
H0
Inc
BF 3
BF 10
BF 30
|package - simulation|
Number of looks



2 *t*-Test

Simulated Design Priors

Analysis Priors

Table 6: Analysis priors used for the t-Test verification run. Designs lists the matching short IDs from the design-prior table.

Function	H1	H0	Designs
tbf01	$\text{Cauchy}_-(0, 0.707)$	Point(0)	T1-T16
tbf01	$\text{Cauchy}_+(0, 0.707)$	Point(0)	T1-T16
tbf01	Cauchy(0, 0.35)	Point(0)	T1-T16

tbf01	Cauchy(0, 0.707)	Point(0)	T1-T16
tbf01	Cauchy(0, 1)	Point(0)	T1-T16
tbf01	StudentT$_+$(3, 0.5, 0.1)	Point(0)	T2-T4, T6, T8, T11, T14-T16
tbf01	StudentT$_+$(3, 0.5, 0.35)	Point(0.2)	T11, T14-T16
tbf01	StudentT$_+$(30, 0.5, 0.35)	Point(0.2)	T1, T5, T7, T10, T13
tbf01	StudentT(3, 0, 0.707)	Point(0)	T1-T16
tbf01	StudentT(30, 0, 0.707)	Point(0)	T1-T16

2.1 Fixed-Design

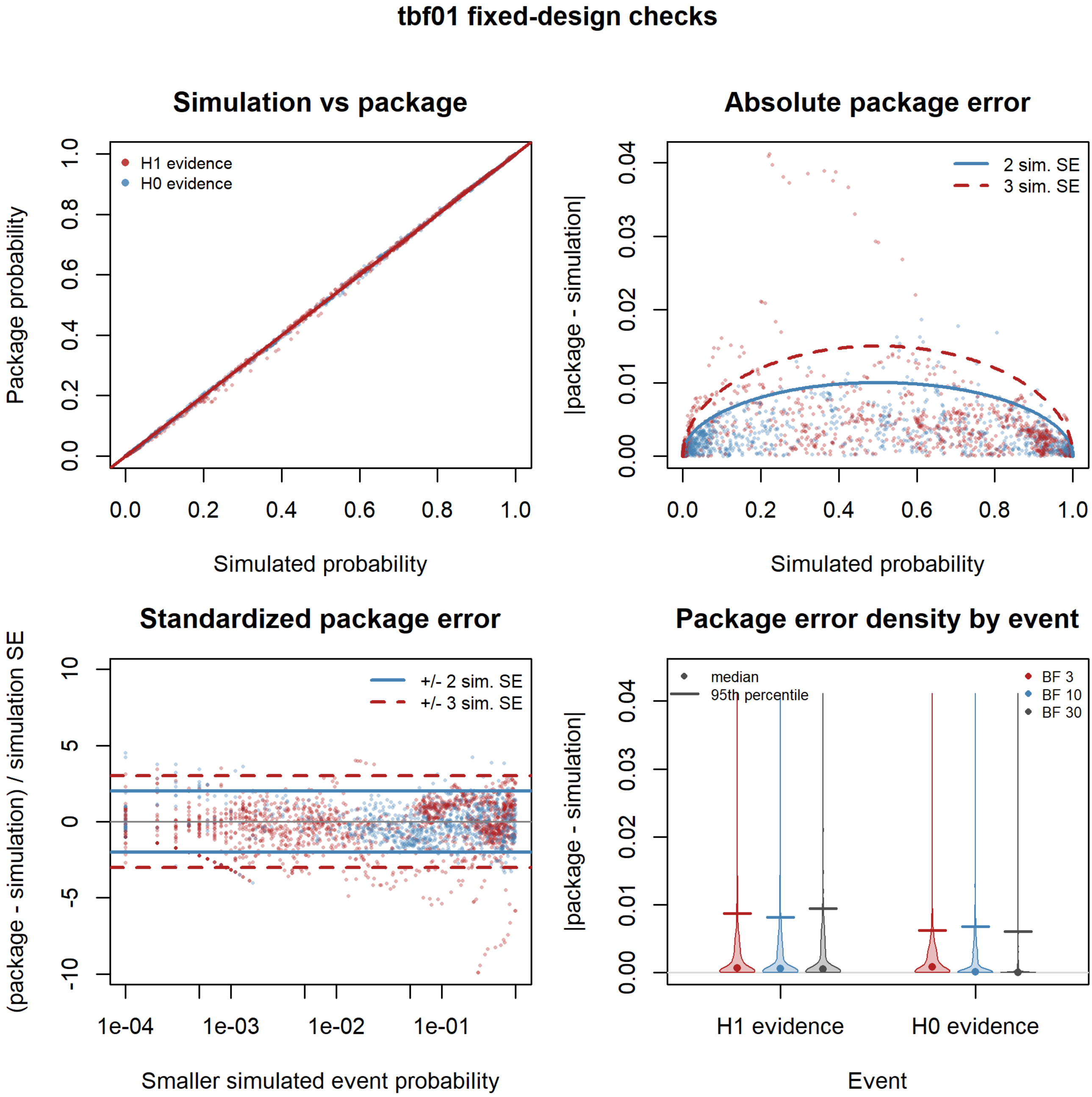
Table 7: Fixed-design verification settings for t-Test analyses.

Function	Analysis priors	Evidence thresholds
tbf01	130	BF 3, 10, 30

The fixed-design plots compare package probability calculations with simulated event frequencies. Some expensive checks are plotted from a precomputed subset; the settings table above describes the full verification design. The panels compare simulated and package probabilities, absolute error, standardized error, and error distributions by event.

Table 5: Simulated design priors for the t-Test verification run. The Design column gives short IDs used by the analysis-prior table.

Design	Design Prior	Sequence	Repetitions	Sampling	n2/n1
T1	Point(0)	10 to 500 by 10	10000	two-sample	1
T2	Point(0.5)	10 to 500 by 10	10000	two-sample	1
T3	Normal(0.5, 0.1)	10 to 500 by 10	10000	two-sample	1
T4	Normal(0.5, 0.5)	10 to 500 by 10	10000	two-sample	1
T5	Point(0)	10 to 500 by 10	10000	one-sample	
T6	Point(0.5)	10 to 500 by 10	10000	one-sample	
T7	Point(0)	10 to 500 by 10	10000	paired	
T8	Point(0.5)	10 to 500 by 10	10000	paired	
T9	Point(-0.4)	10 to 500 by 10	10000	two-sample	1
T10	Point(0)	10 to 500 by 10	10000	two-sample	1.1
T11	Point(0.35)	10 to 500 by 10	10000	two-sample	1.1
T12	Normal(0, 0.5)	10 to 500 by 10	10000	two-sample	1
T13	Point(0)	10 to 10000 by 10	10000	two-sample	1
T14	Point(0.2)	10 to 10000 by 10	10000	two-sample	1
T15	Normal(0.2, 0.1)	10 to 10000 by 10	10000	two-sample	1
T16	Normal(0.2, 0.5)	10 to 10000 by 10	10000	two-sample	1



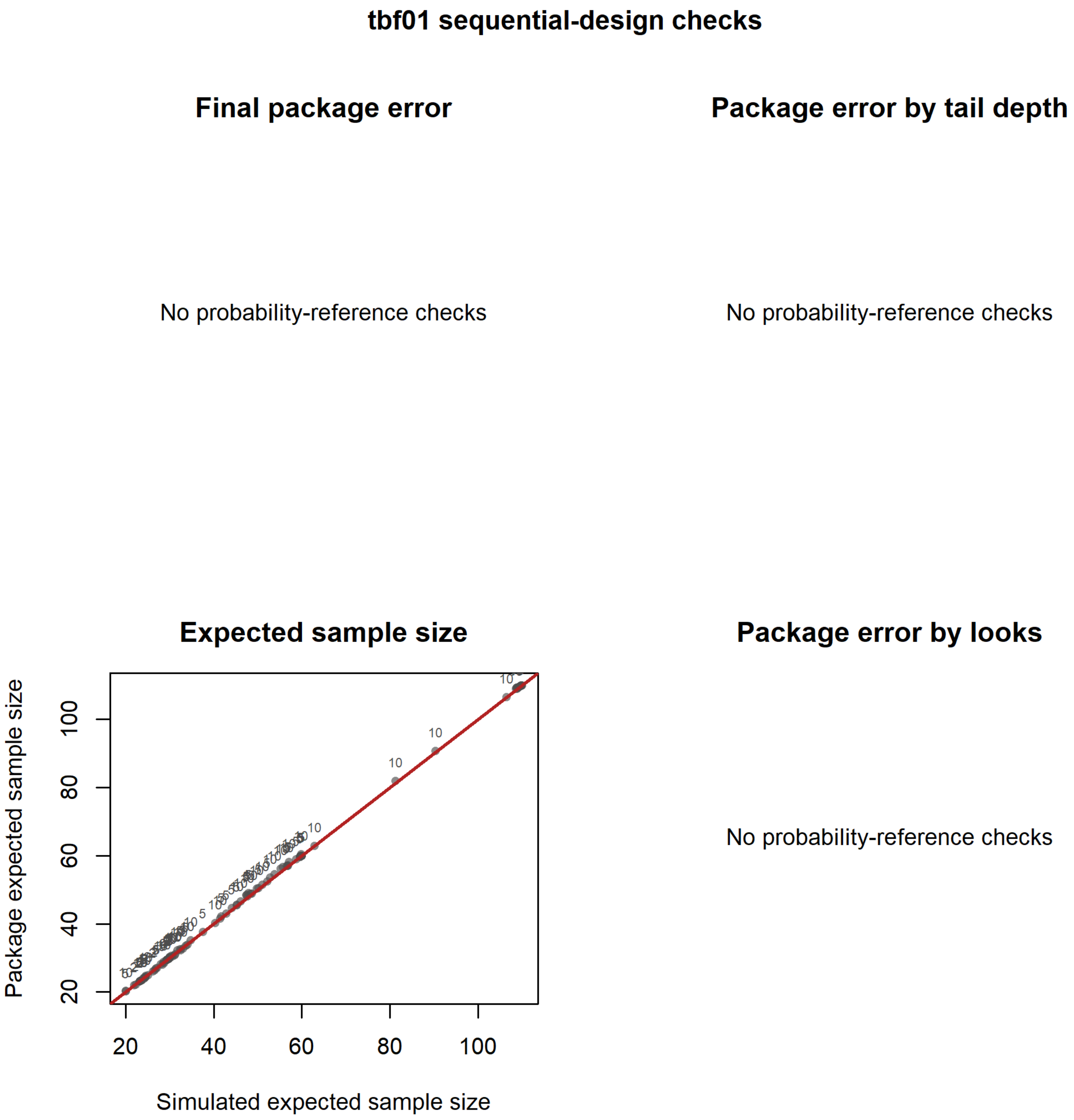
2.2 Sequential-Design

Sequential schedules use looks every 10 observations: one family starts at $n = 20$ and includes 2–20 looks; the stress schedules start at $n = 10$ and include 50, 100, or 200 looks. The 100- and 200-look schedules are compatible with the longer design sequence only.

Table 8: Sequential-design verification settings for t-Test analyses.

Function	Analysis priors	Evidence thresholds	Look sequence
tbf01	130	BF 3, 10, 30	start 20, by 10 (2-20 looks); start 10, by 10 (50, 100, 200 looks)

Sequential-design plots compare expected-sample-size calculations with selected strict t-test simulation settings. Longer schedules are covered by the sample-size search checks below.



3 Binomial Test

Simulated Design Priors

Table 9: Simulated design priors for the Binomial Test verification run. The Design column gives short IDs used by the analysis-prior table.

Design	Design Prior	Sequence	Repetitions
B1	Point(0.5)	10 to 500 by 10	10000
B2	Point(0.6)	10 to 500 by 10	10000
B3	Point(0.2)	10 to 500 by 10	10000
B4	Point(0.75)	10 to 500 by 10	10000
B5	Beta(60, 40)	10 to 500 by 10	10000

B6	Beta(1, 1) on [0.5, 1]	10 to 500 by 10	10000
B7	Beta(1, 1) on [0, 0.5]	10 to 500 by 10	10000
B8	Point(0.5)	10 to 10000 by 10	10000
B9	Point(0.55)	10 to 10000 by 10	10000

Analysis Priors

Table 10: Analysis priors used for the Binomial Test verification run. Designs lists the matching short IDs from the design-prior table.

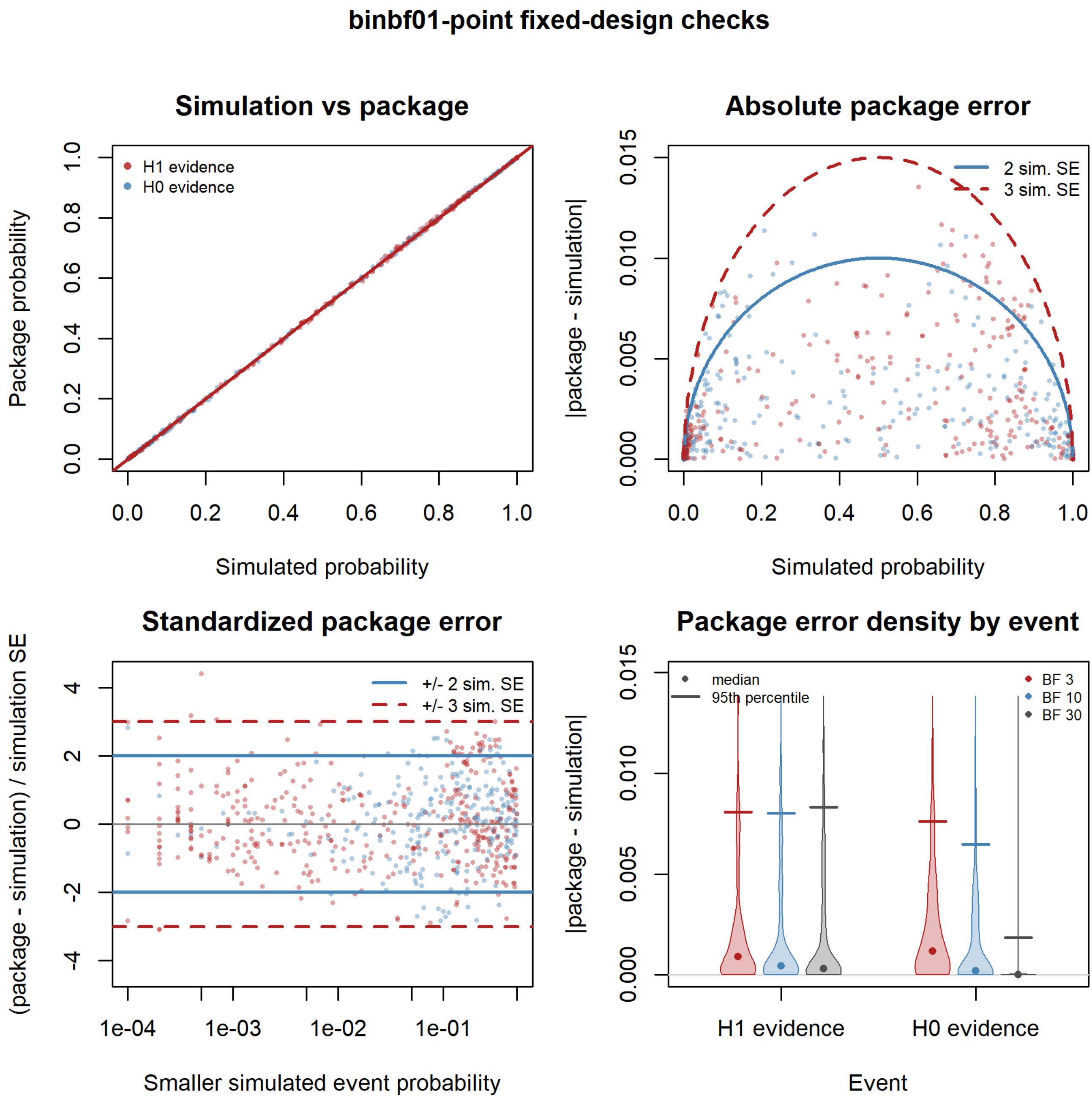
Function	H1	H0	Designs
binbf01 (direction)	$\text{Beta}_+(1,\ 1;\ > 0.2)$	$\text{Beta}_-(1,\ 1;\ \leq 0.2)$	B3
binbf01 (direction)	$\text{Beta}_+(1,\ 1;\ > 0.5)$	$\text{Beta}_-(1,\ 1;\ \leq 0.5)$	B1-B9
binbf01 (direction)	$\text{Beta}_+(1,\ 1;\ > 0.55)$	$\text{Beta}_-(1,\ 1;\ \leq 0.55)$	B9
binbf01 (direction)	$\text{Beta}_+(1,\ 1;\ > 0.6)$	$\text{Beta}_-(1,\ 1;\ \leq 0.6)$	B2, B5
binbf01 (direction)	$\text{Beta}_+(1,\ 1;\ > 0.75)$	$\text{Beta}_-(1,\ 1;\ \leq 0.75)$	B4
binbf01 (point)	Beta(1, 1)	Point(0.2)	B3
binbf01 (point)	Beta(1, 1)	Point(0.5)	B1-B9
binbf01 (point)	Beta(1, 1)	Point(0.55)	B9
binbf01 (point)	Beta(1, 1)	Point(0.6)	B2, B5
binbf01 (point)	Beta(1, 1)	Point(0.75)	B4
binbf01 (point)	Beta(40, 60)	Point(0.5)	B1-B9
binbf01 (point)	Beta(5100, 4900)	Point(0.5)	B8-B9
binbf01 (point)	Beta(60, 40)	Point(0.5)	B1-B9

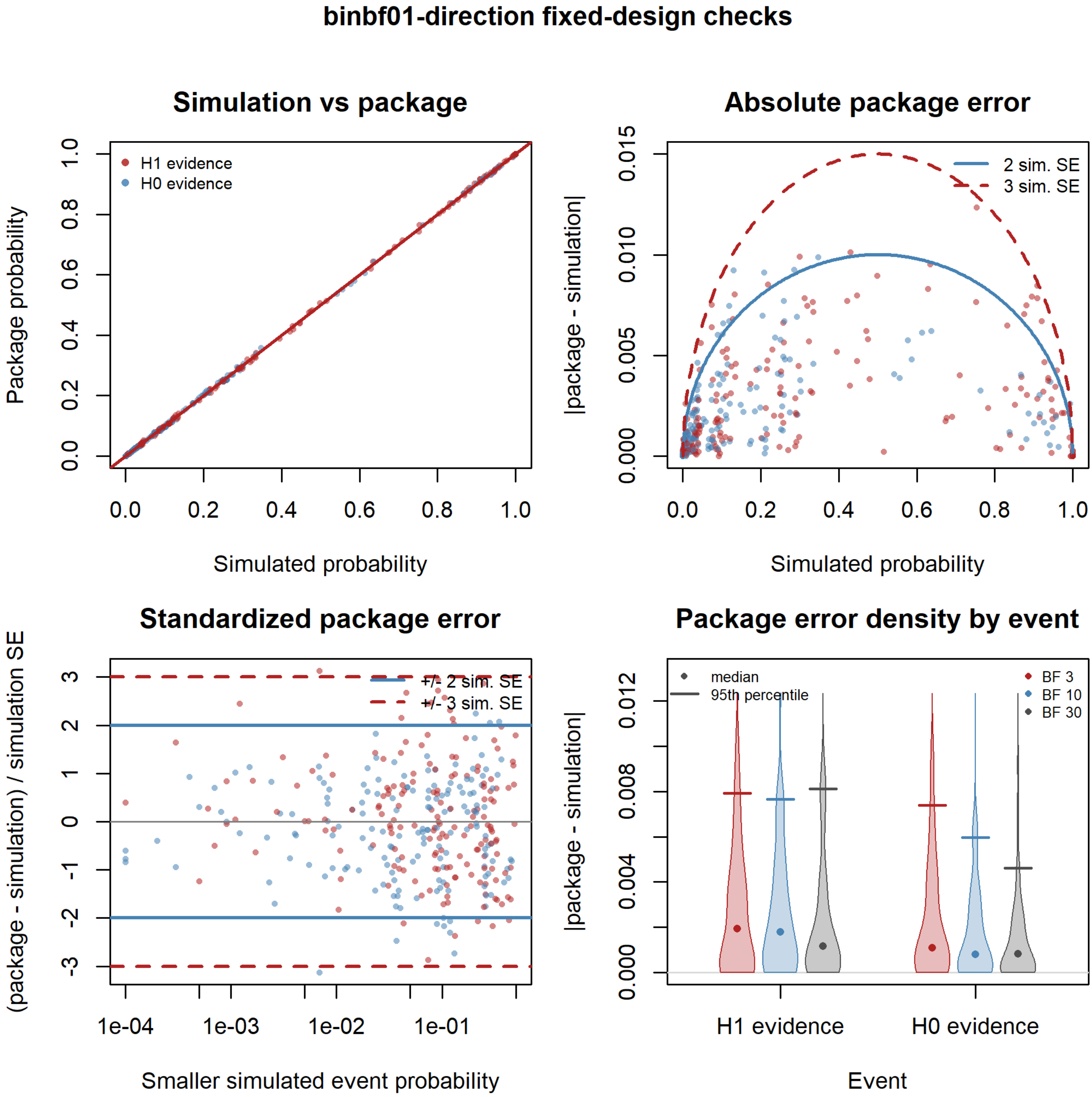
3.1 Fixed-Design

Table 11: Fixed-design verification settings for Binomial Test analyses.

Function	Analysis priors	Evidence thresholds
binbf01 (point)	34	BF 3, 10, 30
binbf01 (direction)	14	BF 3, 10, 30

The fixed-design plots compare package probability calculations with simulated event frequencies. Some expensive checks are plotted from a precomputed subset; the settings table above describes the full verification design. The panels compare simulated and package probabilities, absolute error, standardized error, and error distributions by event.


binbf01-point fixed-design checks
Simulation vs package
H1 evidence
H0 evidence
Package probability
Simulated probability
0.0
0.2
0.4
0.6
0.8
1.0
Absolute package error
2 sim. SE
3 sim. SE
|package - simulation|
0.000
0.005
0.010
0.015
Standardized package error
+/- 2 sim. SE
+/- 3 sim. SE
(package - simulation) / simulation SE
-4
-2
0
2
4
1e-04
1e-03
1e-02
1e-01
Smaller simulated event probability
Package error density by event
median
95th percentile
BF 3
BF 10
BF 30
H1 evidence
H0 evidence
Event



3.2 Sequential-Design

Sequential schedules use looks every 10 observations: one family starts at $n = 20$ and includes 2–20 looks; the stress schedules start at $n = 10$ and include 50, 100, or 200 looks. The 100- and 200-look schedules are compatible with the longer design sequence only.

Table 12: Sequential-design verification settings for Binomial Test analyses.

Function	Analysis priors	Evidence thresholds	Look sequence
binbf01 (point)	34	BF 3, 10, 30	start 20, by 10 (2-20 looks); start 10, by 10 (50, 100, 200 looks)
binbf01 (direction)	14	BF 3, 10, 30	start 20, by 10 (2-20 looks); start 10, by 10 (50, 100, 200 looks)

Sequential binomial simulation settings are retained for future package-function validation. The package does not currently expose a sequential binomial evaluation function, so no sequential binomial package-calculation figure is rendered.

4 Package Checks

Package checks are selected from the simulation corpus and recomputed when this report is built. The broader corpus contains the stochastic simulation output; the release checks define which current package calls are compared against that output. Fixed-design rows check probability functions. Sequential rows check probability, expected sample size, and selected sample-size search behavior.

Table 13: Package checkout used for recomputed verification results.

Package version	Git revision	Package tree dirty	Recomputed
0.3	6240269c7a08	no	2026-07-19 16:20:02 CEST

Table 14: Package check coverage. Rows are explicit release verification cases, not PDF-side selections.

Role	Family	Mode	Function	Rows	Required	Simulation sets	Designs	Priors	Schedules	BF thresholds
Fixed probability checks	binomial	fixed	pbinbf01	1,440	1,440	2	9	48	0	3, 10, 30
Fixed probability checks	t	fixed	ptbf01	3,900	3,900	1	16	130	0	3, 10, 30
Fixed probability checks	z	fixed	pbf01	2,880	2,880	2	15	96	0	3, 10, 30
Fixed probability checks	z	fixed	pnmbf01	1,050	1,050	1	15	35	0	3, 10, 30
Sequential probability and expected sample size checks	z	sequential	pbf01seq	216	216	3	8	43	3	3, 10, 30
Sequential expected sample size checks	t	sequential	ptbf01seq	144	144	1	3	24	3	3, 10, 30
Sequential sample-size search	t	sequential	ntbf01seq	144	26	1	11	76	8	3, 10, 30
Sequential sample-size search	z	sequential	nbf01seq	144	50	3	12	68	8	3, 10, 30
Simulation data checks	binomial	sequential	(none)	2	2	2	0	0	0	

5 Sample-Size Search Checks

These checks compare package sample-size searches with sample-size targets observed in the simulations. Primary rows are settings where the simulation clearly reached the requested probability. Exploratory rows record settings where the simulated range did not clearly reach the target; they are shown for context and are not treated as package pass/fail checks.

Table 15: Sample-size search overview for the rendered corpus snapshot.

Generated	Policy	Comparison rows	Package rows	Comparable rows	Sequential search cases	Recomputed rows	Recompute time	Exploratory rows run
2026-07-19 16:24:51 CEST	Standard checks plus release cases	19,080	76	76	288	76	15.92 s	no

Table 16: Sequential sample-size search coverage. Primary rows are evaluated by the package; exploratory rows document simulation cases that did not clearly reach the target.

Function	Role	Simulation status	BF types	Evidence	BF thresholds	Targets	Rows	Reached	Certified	Errors	Schedules	Designs	Priors	n range
nbf01seq	Primary search checks	Reached in simulation	directional, moment, normal	H0, H1	3, 10, 30	0.1, 0.3, 0.8, 0.95	50	50	50	0	7	10	33	10-770
nbf01seq	Exploratory search checks	Reached in simulation	moment, normal	H0, H1	3, 10, 30	0.1, 0.3, 0.8	7	0	0	0	5	5	7	20-170
nbf01seq	Exploratory search checks	Not reached by simulated range	directional, moment, normal	H0, H1	3, 10, 30	0.1, 0.3, 0.8, 0.95	47	0	0	0	7	10	40	
nbf01seq	Exploratory search checks	Not reached in simulation	directional, moment, normal	H0, H1	3, 10, 30	0.1, 0.3, 0.8, 0.95	40	0	0	0	6	10	26	
ntbf01seq	Primary search checks	Reached in simulation	t	H0, H1	3, 10, 30	0.1, 0.3, 0.8, 0.95	26	26	26	0	7	10	23	20-280
ntbf01seq	Exploratory search checks	Reached in simulation	t	H0, H1	3, 10, 30	0.1, 0.3, 0.8, 0.95	12	0	0	0	7	7	12	30-910
ntbf01seq	Exploratory search checks	Not reached by simulated range	t	H0, H1	3, 10, 30	0.1, 0.3, 0.8, 0.95	46	0	0	0	7	10	39	
ntbf01seq	Exploratory search checks	Not reached in simulation	t	H0, H1	3, 10, 30	0.1, 0.3, 0.8, 0.95	60	0	0	0	8	11	43	

Table 17: Package-evaluated sequential sample-size search rows. Only rows sent to the package search are counted; fractional n error compares package and simulation first crossing sample sizes.

Family	BF type	Search profile	Evaluated rows	Reached	Certified	Errors	n differs	P90 abs fractional n error	Max abs fractional n error
t	t	seq-start10-by10	9	9	9	0	0	0	0
t	t	seq-start20-by10	17	17	17	0	1	0	0.5
z	directional	seq-start10-by10	4	4	4	0	0	0	0
z	directional	seq-start20-by10	11	11	11	0	0	0	0
z	moment	seq-start10-by10	6	6	6	0	0	0	0
z	moment	seq-start20-by10	14	14	14	0	1	0	0.1
z	normal	seq-start10-by10	5	5	5	0	0	0	0
z	normal	seq-start20-by10	10	10	10	0	1	0.013	0.125

Table 18: Sample-size search notes.

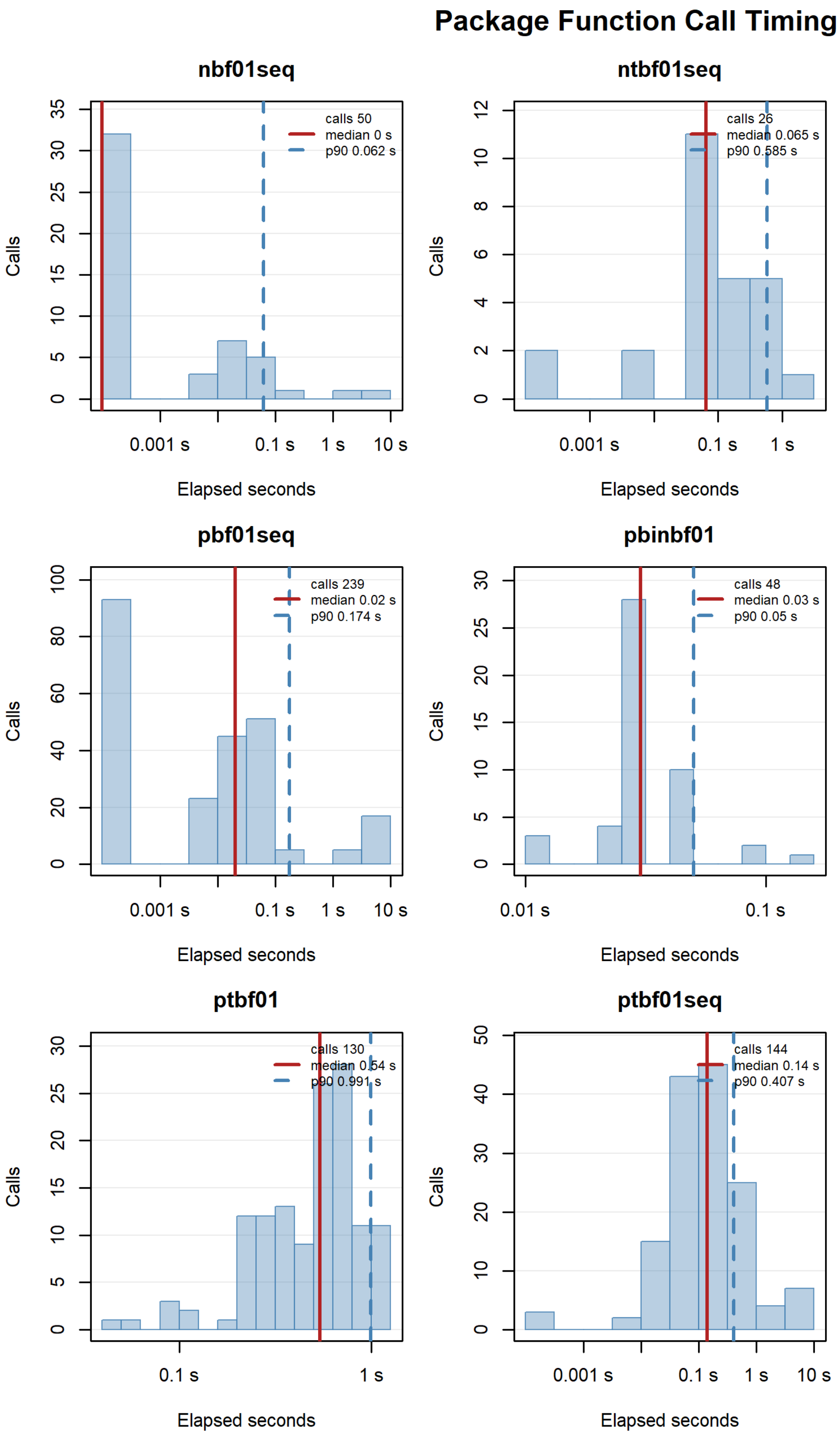
Severity	Note	Rows	Details
Note	Exploratory rows need review	100	Some exploratory rows did not reach the requested probability within the simulated range; they

6 Run Time Diagnostics

The build records elapsed time for package calls made while recomputing the checks. Fixed-design probability calls can cover several rows at once because those functions are vectorized; most sequential and sample-size search calls are one setting at a time.

Table 19: Package call run time summary. Fixed probability rows can be vectorized, so calls and rows covered need not match.

Source	Function	Calls	Rows covered	Median	P90	Max	Total
reference	pbf01	73	2,190	0 s	0.01 s	0.04 s	0.19 s
reference	pbf01seq	239	906	0.02 s	0.174 s	5.91 s	1.68 min
reference	pbinbf01	48	1,440	0.03 s	0.05 s	0.14 s	1.69 s
reference	pnmbf01	35	1,050	0 s	0.016 s	0.11 s	0.19 s
reference	ptbf01	130	3,900	0.54 s	0.991 s	1.11 s	1.197 min
reference	ptbf01seq	144	144	0.14 s	0.407 s	6.23 s	1.148 min
search	nbf01seq	50	50	0 s	0.062 s	7.75 s	9.48 s
search	ntbf01seq	26	26	0.065 s	0.585 s	1.5 s	5.62 s



7 Summary

The tables and plots in this report are a compact summary of the simulation verification suite. The broad Monte Carlo corpus defines the available simulation evidence, while the release checks define the package function calls recomputed for this report. The automated package checks remain the authoritative pass/fail record for development.